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BLOBFILTERS

A MATLAB TOOLBOX FOR MITOCHONDRIAL STRUCTURE ENHANCEMENT

Contents

1 *Introduction* 5

1.1 *Quick start* 5

1.2 *Image conventions* 6

1.3 *Parameter defaults* 6

2 *Blob and Puncta Detectors* 7

2.1 `logEnhance` 7

2.2 `granulometryEnhance` 7

3 *Rod and Fibre Detectors* 9

3.1 `fiberEnhance` 9

3.2 `capsuleEnhance` 9

3.3 `rodGranulometryEnhance` 10

3.4 `cellposeEnhance` 10

4 *Orientation Analysis and Pre-filtering* 19

4.1 `structureTensorEnhance` 19

4.2 `orientedGaussSmooth` 20

5 *Demo Scripts* 21

5.1 `demoMitoEnhance` 21

5.2 `demoPrefilter` 21

5.3 `demoMitoEnhance` — *Cellpose comparison* 25

6 *Unit Tests* 29

References 31

7 *Bibliography* 33

1
Introduction

BlobFilters is a MATLAB toolbox for enhancing elongated bright structures—rods, fibres and tubular networks—against a noisy fluorescence background. Its primary application is the detection and segmentation of mitochondria in confocal images, where individual organelles appear as either compact *puncta* or elongated *rods and networks* of approximately 8 px width and 12–40 px length.

The toolbox provides seven functions organised into three tiers:

Tier	Functions
Blob / puncta detectors	`logEnhance`, `granulometryEnhance`
Rod / fibre detectors	`fiberEnhance`, `capsuleEnhance`, `rodGranulometryEnhance`
Orientation and pre-filtering	`structureTensorEnhance`, `orientedGaussSmooth`

1.1 Quick start

Add `src/` and `demos/` to the MATLAB path, then:

```
addpath src demos

I = im2single(imread('my_confocal.tif'));

% Rod detector
pFib.widths  = [6 7 8 9 10];
R_fiber      = fiberEnhance(I, pFib);

% Rod granulometry
pRod.lengths = [8 12 16 20 28 36];
R_rod        = rodGranulometryEnhance(I, pRod);

% Structure tensor coherence as post-weight
pST.sigmaGrad = 1.5; pST.sigmaInt = 5;
C            = structureTensorEnhance(I, pST);
R_weighted   = R_rod .* C;    % suppress puncta
```

1.2 *Image conventions*

All functions accept any MATLAB numeric type and convert internally to `single` in $[0, 1]$ via `im2single`. Inputs are expected to be 2-D (single-channel) grayscale images; passing an RGB array raises an error.

1.3 *Parameter defaults*

All parameters have sensible defaults tuned to 8 px-wide, 12–40 px-long mitochondria imaged at typical confocal resolution. Pass an empty struct or omit the second argument to use defaults:

```
R = fiberEnhance(I);           % use all defaults
R = fiberEnhance(I, struct()); % equivalent
```

2
Blob and Puncta Detectors

2.1 `logEnhance`

The Laplacian-of-Gaussian (LoG) is the canonical blob detector in scale-space theory [1]. At scale σ the normalised LoG response is

[1]; and

$$R_\sigma = -\sigma^2 \nabla^2 \big(G_\sigma * I\big), \tag{2.1}$$

where G_σ is a Gaussian with standard deviation σ and $\nabla^2 = \partial_{xx} + \partial_{yy}$ is the Laplacian. The negative sign gives a strong *positive* response at bright circular blobs whose characteristic radius is approximately $\sqrt{2}\,\sigma$.

`logEnhance` evaluates (2.1) at each requested scale and returns the per-pixel maximum:

$$R = \max_{\sigma \in \Sigma} R_\sigma. \tag{2.2}$$

Separable implementation

Rather than convolving directly with a 2-D LoG kernel, the image is first smoothed by two 1-D Gaussian passes[2] and then filtered with the compact 3×3 discrete Laplacian $\ell = [0, 1, 0;\ 1, -4, 1;\ 0, 1, 0]$. This reduces the per-scale cost from $O(N^2 (6\sigma)^2)$ to $O(N^2 \cdot 6\sigma)$, approximately a $6\times$ speedup at $\sigma = 6$.

[2] `imfilter` requires the kernel to be of class `double` regardless of the image class; both the 1-D Gaussian and the Laplacian are kept as `double`.

Parameters

```
pLog.sigmas    = [2 3 4 5 6];   % sigma ~ radius/sqrt(2)
pLog.normalize = true;
R = logEnhance(I, pLog);
```

`sigmas`
Vector of σ values (px). Default `[2 3 4 5 6]`.
Rule of thumb: $\sigma \approx r/\sqrt{2}$ for a blob of radius r.

`normalize`
Rescale output to $[0, 1]$. Default `true`.

2.2 `granulometryEnhance`

Granulometry characterises the size distribution of bright objects via the *pattern spectrum* [3]. A morphological opening with a disk of radius r removes all bright structures narrower than $2r$; the residue

[3]; and

$$\mathrm{PS}(r) = \mathrm{Open}(I, \mathrm{disk}(r)) - \mathrm{Open}(I, \mathrm{disk}(r+\Delta r)) \tag{2.3}$$

is large wherever structures of scale r exist. Summing over all consecutive radius pairs yields an integrated enhancement map that highlights objects across all specified scales.

Disk openings are *isotropic*: compact puncta and short rods at the same scale both respond. Multiplying the output by the structure tensor coherence C (Section 4.1) suppresses round puncta.

Parameters

```
pGran.sigmas    = [2 4 6 8 10 12 16];
pGran.normalize = true;
G = granulometryEnhance(I, pGran);
```

`sigmas`
Disk radii (px). Default `[1 2 3 4 8 16 32]`.
At least 2 values are required.

`normalize`
Default `true`.

3
Rod and Fibre Detectors

3.1 `fiberEnhance`

FIBERENHANCE wraps the MATLAB Image Processing Toolbox function `fibermetric`, which implements a Frangi-style tubularity measure [1]. At each scale w the Hessian of the Gaussian-smoothed image is computed; for a bright tubular ridge the smaller eigenvalue λ_1 is large and negative while $\lambda_2 \approx 0$. The per-pixel response is

[1]

$$S_w = e^{-\lambda_1}\left(1 - e^{\lambda_2/c}\right), \tag{3.1}$$

where c is an automatic scale factor. `fibermetric` aggregates over all widths and selects the best-scale response per pixel.

Multimode options

`'builtin'` (default) Passes all widths to `fibermetric` at once; MathWorks selects the best scale per pixel.

`'stack'` Calls `fibermetric` once per width and takes the pixel-wise maximum—useful for inspecting individual scale responses.

Parameters

```
pFib.widths    = [6 7 8 9 10];
pFib.multimode = 'builtin';
pFib.normalize = true;
R = fiberEnhance(I, pFib);
```

`widths`
Cross-sectional widths (px). Default `[6 7 8 9 10]`.
Probe the *short axis* only; length is not a parameter.

`multimode`
`'builtin'` or `'stack'`. Default `'builtin'`.

Requires IPT R2018b+.

3.2 `capsuleEnhance`

CAPSULE KERNELS are oriented templates shaped like a stadium[2]—a rectangle capped with two semicircles—that match the cross-sectional profile of a mitochondrion. A bank of kernels at different lengths and orientations is convolved with the image, and the per-pixel maximum response over the bank is returned.

Two kernel modes are available:

[2] A *stadium* (or *discorectangle*) is a rectangle capped by two half-disks of the same radius. Also known as an oblong or a capsule.

'single' Normalised positive capsule kernel. Gives high response wherever a bright rod matches the kernel footprint.

'doc' Difference-of-Capsules (DoC): the response of a narrow inner capsule minus α times a wider inhibitory surround. Suppresses signal from clustered or touching mitochondria.

Kernels are unit-sum-normalised after mean subtraction so that a flat background gives zero response.

Parameters

```
% Single mode
pCapS.lengths      = [12 16 20 28 36 40];
pCapS.width        = 8;
pCapS.orientations = 12;
pCapS.mode         = 'single';
R_single = capsuleEnhance(I, pCapS);

% DoC mode
pCapD.wideWidth = 18; pCapD.alpha = 0.55;
pCapD.mode      = 'doc';
R_doc = capsuleEnhance(I, pCapD);
```

lengths
Capsule body lengths (px). Default `[12 16 20 28 36 40]`.

width
Capsule half-width / end radius (px). Default `8`.

wideWidth
DoC outer radius (px). Default `16`.

alpha
DoC inhibitory weight $\in (0, 1)$. Default `0.5`.

orientations
Number of angles in $[0°, 180°)$. Default `12`.

mode
`'single'` or `'doc'`. Default `'single'`.

3.3 `rodGranulometryEnhance`

Rod granulometry replaces the isotropic disk SE of `granulometryEnhance` with a bank of oriented line structuring elements [3]. At each length L the maximum opening over all orientations θ is computed:

[3]

$$O_{\max}(L) = \max_{\theta} \operatorname{Open}(I, \operatorname{line}(L, \theta)), \tag{3.2}$$

and the pattern spectrum residue $O_{\max}(L_k) - O_{\max}(L_{k+1})$ isolates structures of length approximately L_k pixels.

Because a compact punctum cannot survive a line opening longer than its own diameter, line SEs are *inherently* selective for elongated structures without needing coherence post-weighting—a key advantage over disk granulometry.

Parameters

```
pRod.lengths      = [8 12 16 20 28 36];
pRod.orientations = 8;
R = rodGranulometryEnhance(I, pRod);
```

lengths
Line lengths (px). Default `[8 12 16 20 28 36]`.
Rod of length L_{rod} responds at the largest $L_k < L_{\text{rod}}$.

orientations
Number of angles in $[0°, 180°)$. Default `8`.

normalize Default `true`.

3.4 `cellposeEnhance`

CELLPOSEENHANCE wraps the MathWorks Medical Imaging Toolbox Interface for Cellpose Library [4], providing deep-learning cell segmentation from within MATLAB.

[4]; and

Unlike the other BlobFilters enhancers, which return continuous probability-like maps, `cellposeEnhance` returns a *binary* mask and an integer label image:

R Binary single $\{0, 1\}$ mask compatible with the $[0, 1]$ convention of every other enhancer. Can be used as a multiplicative weight, a seed mask, or a segmentation result directly.

L `uint16` label image. Each detected object carries a unique positive integer; background is 0. Neighbouring or touching objects that would merge under simple thresholding are distinguished here.

```
[R, L] = cellposeEnhance(I);          % defaults: cyto3,
    diameter 10 px
imshow(label2rgb(L, 'hsv', 'k'))      % coloured instance map
```

Parameters

Model choice for mitochondria

The `'cyto3'` model (Cellpose 3, Nature Methods 2025) is a suitable default for fluorescence mitochondria images: it was trained on nine heterogeneous datasets and generalises well to sub-cellular structures. For higher specificity, fine-tune a custom model with 100–200 manually annotated frames using the Cellpose 2.0 human-in-the-loop GUI, then supply the saved model path:

```
pCP.model    = 'C:\models\mito_finetuned';
pCP.diameter = 10;
[R, L] = cellposeEnhance(I, pCP);
```

`model`
Cellpose model name or path to a custom model file. Default `'cyto3'`.

`diameter`
Expected object cross-section (px). Default `10`.
Set to `0` for automatic estimation (slower).

Requires the *Medical Imaging Toolbox Interface for Cellpose Library* add-on and a Python environment with `cellpose` installed.

Parameter optimisation for mitochondria fluorescence

A SYSTEMATIC SWEEP of `cellProb` and `flowThreshold` was carried out using the companion script `sweepCellpose.m` (located in `Segmentation_sandbox/demos/`) on a cropped fluorescence mitochondria image (`mitImagecrop.mat`, same field of view used throughout). Two additional axes were explored: the specialised `bact_fluor_cp3` model (Cellpose 3, trained on fluorescent bacteria whose morphology resembles elongated mitochondria) and an increased iteration count (`nIter` = 2000, recommended in the literature for long thin structures [5]).

[5]

Parameters swept.

- `cellProb` (`CellThreshold`): broad survey $[-4, \ldots, +2]$, narrowed to $[-2, -1, 0, +1]$, then finalised at $[-1, 0, +1]$.
- `flowThreshold` (`FlowErrorThreshold`): $[0.30, 0.50, 0.80, 1.20]$, finalised at $[0.60, 0.80, 1.00]$.

Metrics
n objects: number of distinct detected objects (`max(L(:))`); too high indicates over-segmentation or noise detections.

coverage %: fraction of image pixels labelled as foreground; should match the expected mitochondrial area fraction.

- Models: `cyto3` and `bact_fluor_cp3`.
- `nIter`: 0 (default ≈200) and 2000.

Findings.

- `bact_fluor_cp3` produced over-segmentation on this image and performed worse than `cyto3` across all parameter combinations.
- Increasing `nIter` to 2000 gave no improvement in object count, coverage, or visual boundary quality for `cyto3`.
- The useful operating region for `cyto3` was `cellProb` $\in [-1, 0]$ combined with `flowThreshold` $\in [0.8, 1.2]$. Outside this region object counts increased sharply (over-segmentation) or dropped steeply (under-detection).
- `cellProb` $= 0$ produced visually tighter outlines that better matched individual mitochondria compared with the more permissive -1.

Results (n objects / coverage %, `minSize` = 64 px^2).

cp \ ft	0.60	0.80	1.00
−1	21 / 7.4%	21 / 7.4%	21 / 7.4%
0	21 / 5.9%	21 / 5.9%	21 / 5.9%
+1	21 / 4.6%	21 / 4.6%	21 / 4.6%

Table 3.1: `cyto3`, default `nIter`.

cp \ ft	0.60	0.80	1.00
−1	22 / 7.2%	22 / 7.2%	22 / 7.2%
0	20 / 5.4%	21 / 5.5%	21 / 5.5%
+1	17 / 3.4%	19 / 3.5%	19 / 3.5%

Table 3.2: `cyto3`, `nIter` = 2000.

cp \ ft	0.60	0.80	1.00
−1	31 / 11.7%	32 / 11.9%	32 / 11.9%
0	30 / 11.0%	31 / 11.2%	31 / 11.2%
+1	31 / 10.4%	31 / 10.4%	31 / 10.4%

Table 3.3: `bact_fluor_cp3`, default `nIter`.

Recommended settings (used as ground truth in `sweepSegmentation.m`):

```
pCP.model         = 'cyto3';
pCP.diameter      = 10;      % px - typical mitochondrion cross
    -section
pCP.cellProb      = 0;       % CellThreshold; lower = more
    sensitive
pCP.flowThreshold = 0.8;     % FlowErrorThreshold
[R, L] = cellposeEnhance(I, pCP);
```

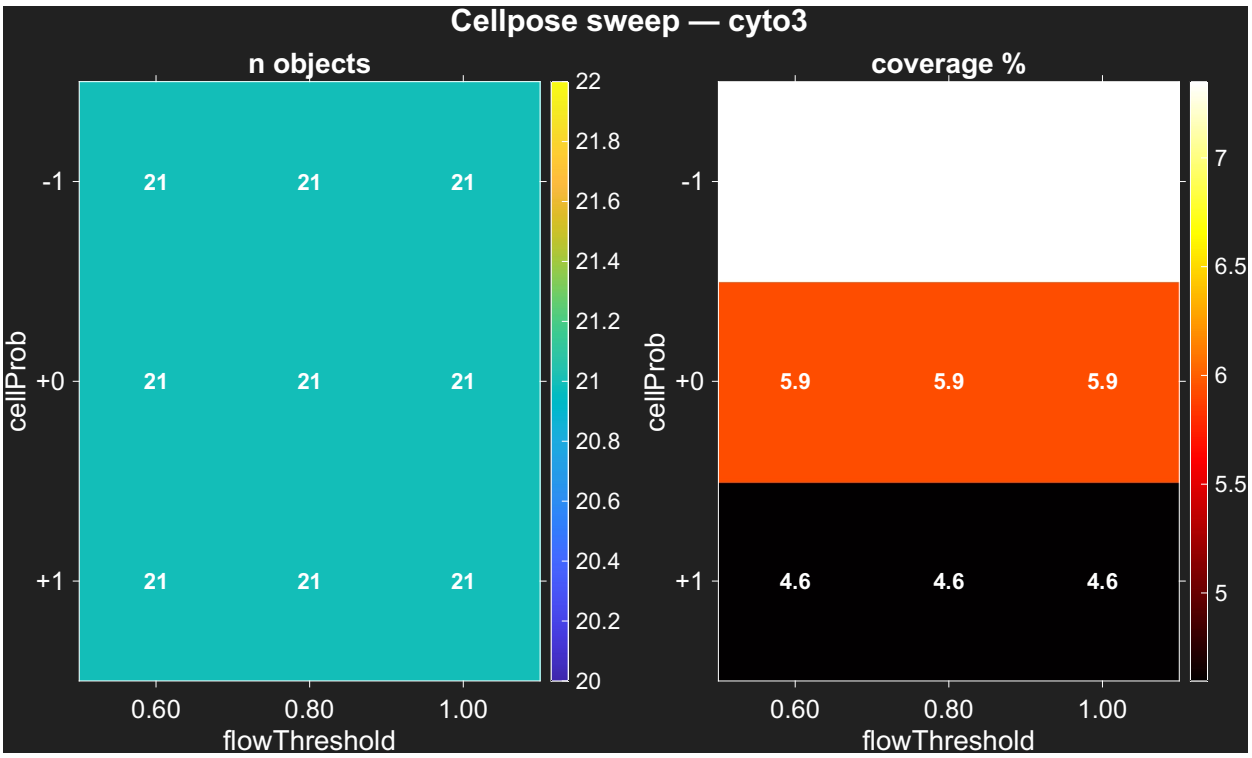

Figure 3.1: `cyto3` sweep heatmaps (default `nIter`). Left: number of detected objects. Right: foreground coverage (%). Rows = `cellProb` $\in \{-1, 0, +1\}$; columns = `flowThreshold` $\in \{0.60, 0.80, 1.00\}$. The cell at `cellProb` $= 0$, `flowThreshold` $= 0.8$ gives the best balance of object count and boundary tightness.

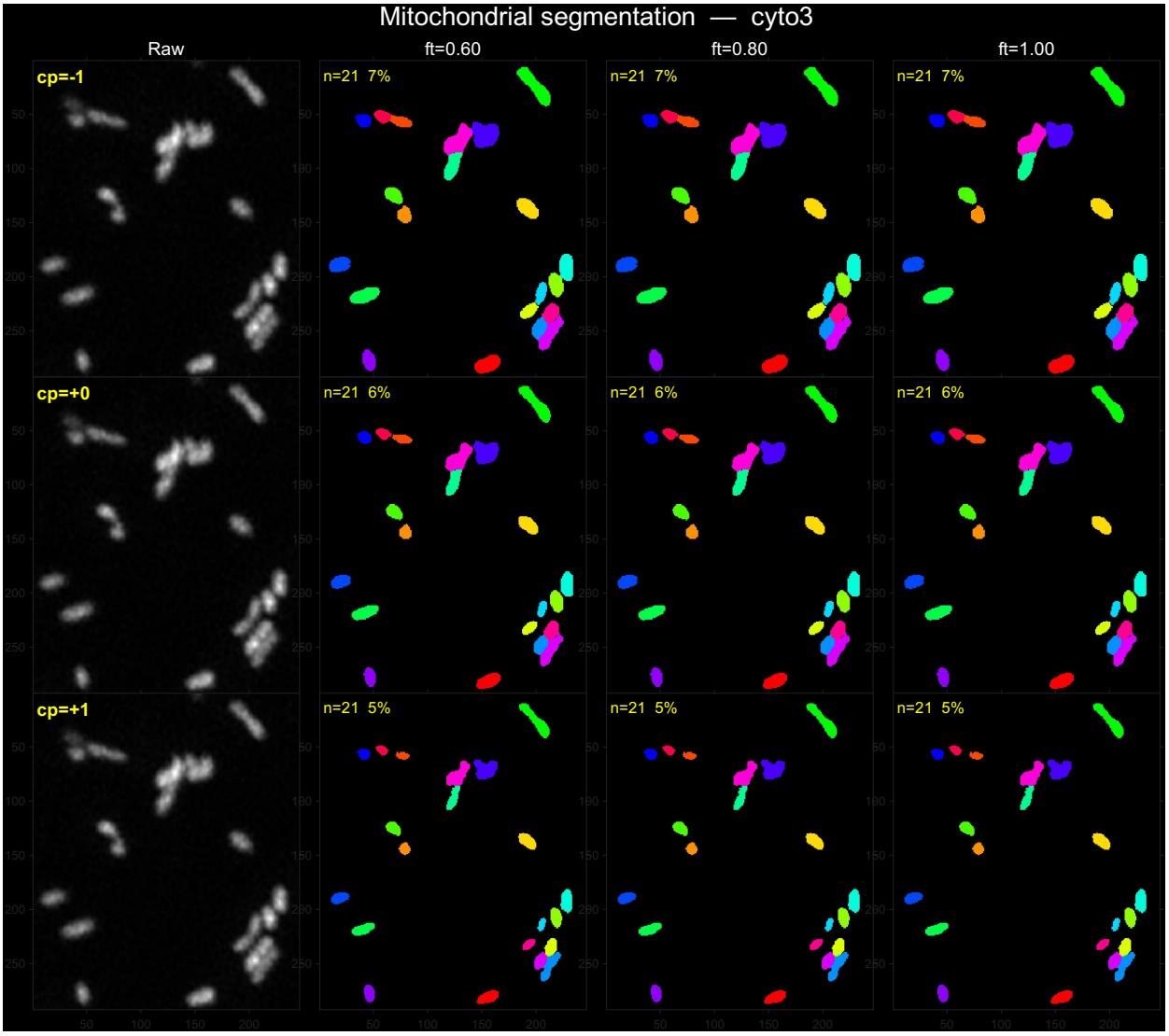

Figure 3.2: `cyto3` label grid (default `nIter`). Rows = `cellProb` (-1, 0, $+1$). Column 1: raw image. Columns 2–4: instance labels (`label2rgb`) for `flowThreshold` = 0.60, 0.80, 1.00. Black pixels are background.

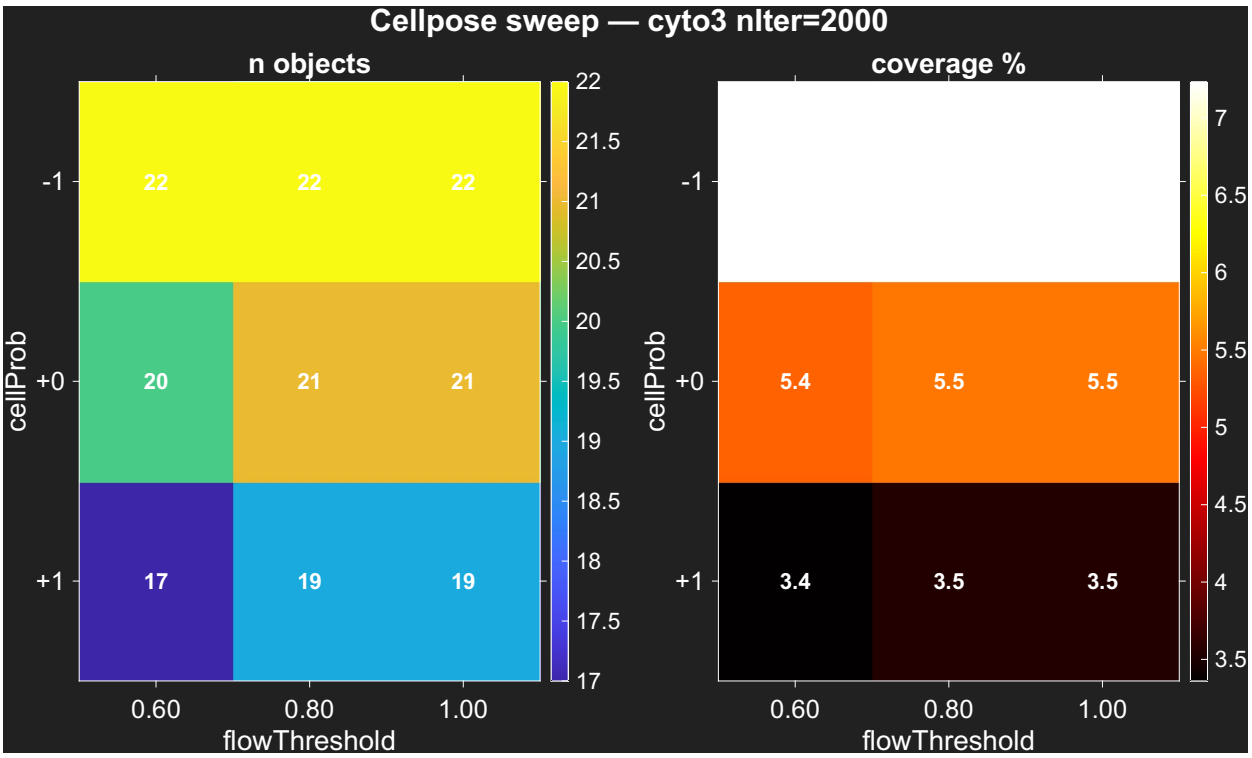

Figure 3.3: `cyto3` sweep heatmaps with `nIter = 2000`. Layout as Figure 3.1. Compare with the default-`nIter` result to assess whether additional iterations alter object count or coverage.

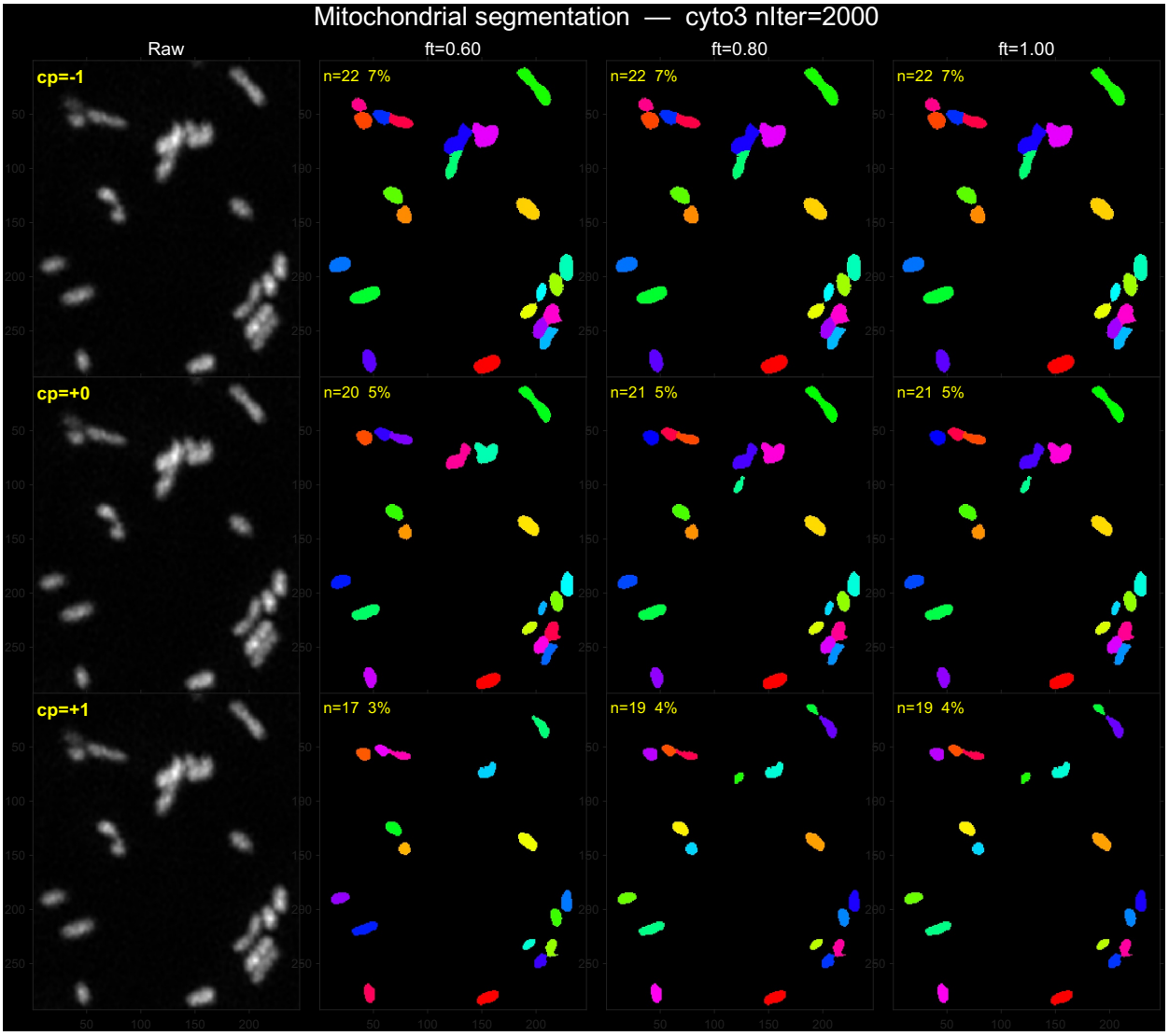

Figure 3.4: `cyto3` label grid with `nIter` = 2000. Layout as Figure 3.2.

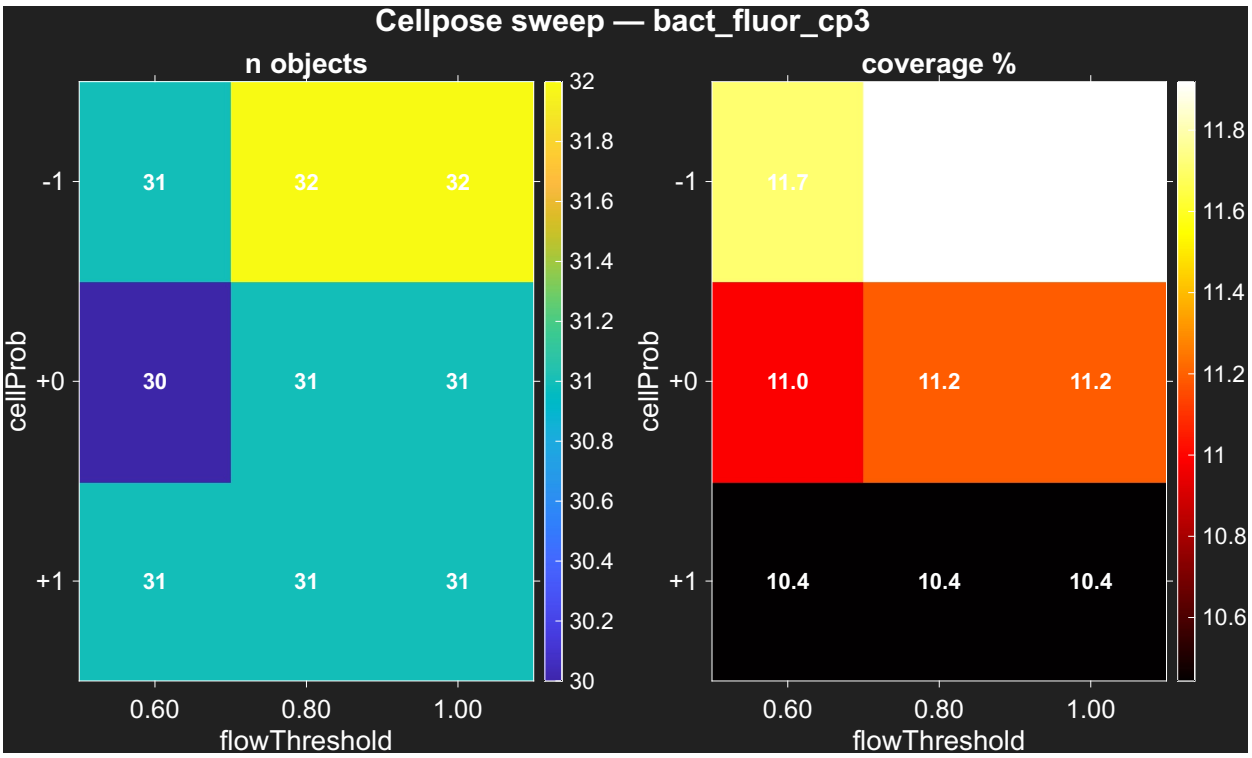

Figure 3.5: `bact_fluor_cp3` sweep heatmaps (default `nIter`). Layout as Figure 3.1. This model is trained on fluorescent bacteria; compare object counts and coverage with the `cyto3` results above.

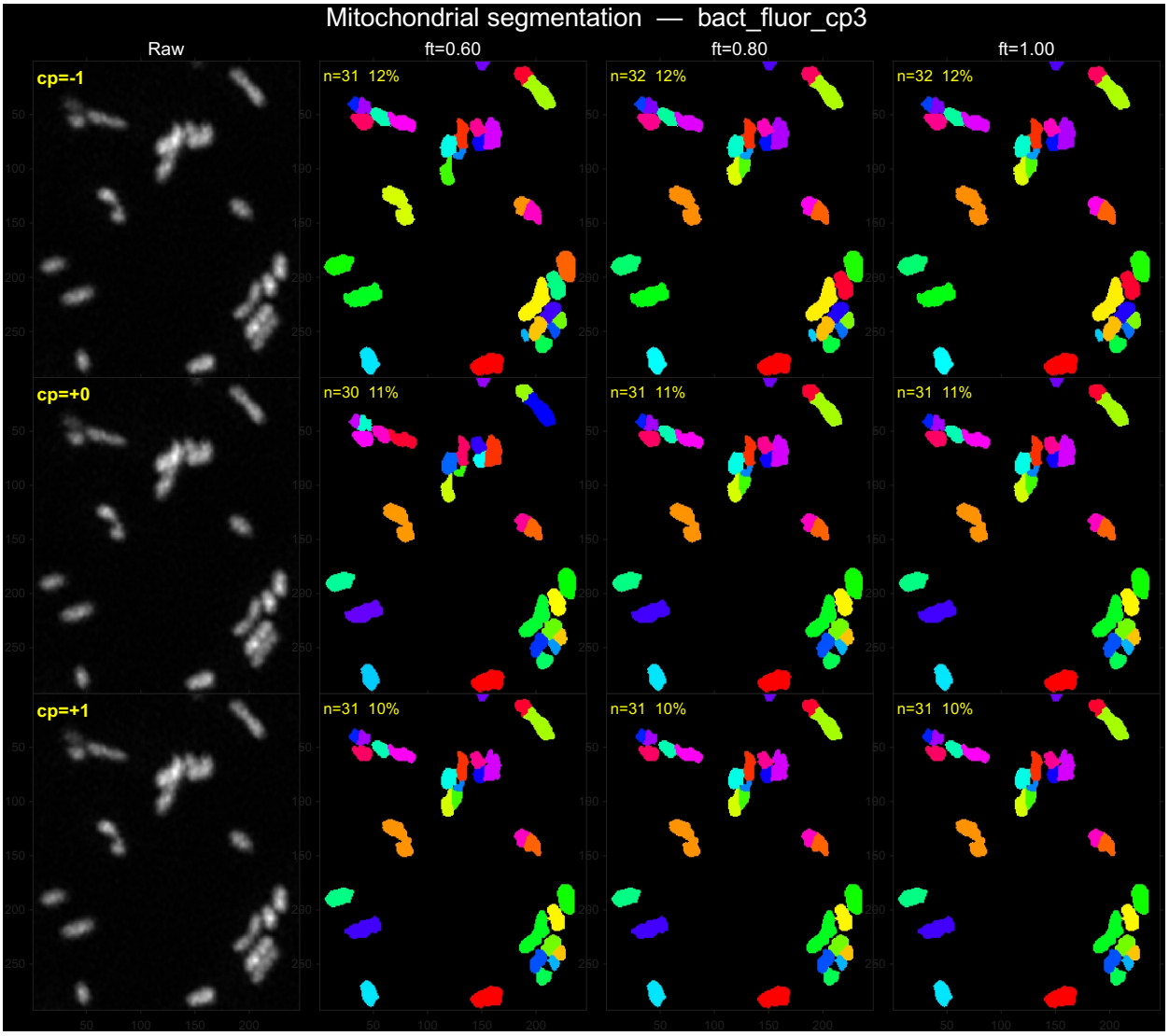

Figure 3.6: `bact_fluor_cp3` label grid (default `nIter`). Layout as Figure 3.2.

4
Orientation Analysis and Pre-filtering

4.1 `structureTensorEnhance`

THE STRUCTURE TENSOR [1] encodes local image orientation. The tensor is formed from the outer product of the image gradient, smoothed at an integration scale:

[1]; and

$$J = G_{\sigma_i} * \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}, \tag{4.1}$$

where G_{σ_i} denotes convolution with a Gaussian of standard deviation σ_i (the integration scale). The eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$ of J satisfy:

- $\lambda_1 \approx \lambda_2 \approx 0$: flat region (no edges)
- $\lambda_1 \gg \lambda_2 \approx 0$: linear edge (rod or fibre)
- $\lambda_1 \approx \lambda_2 \gg 0$: corner or isotropic texture

The *coherence index*

$$C = \left(\frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}\right)^2 \in [0, 1] \tag{4.2}$$

is 1 for a perfect linear structure and 0 for an isotropic region. Eigenvalues are computed from the analytically stable formula $\lambda_{1,2} = \frac{1}{2}\left[(J_{11} + J_{22}) \pm \sqrt{(J_{11} - J_{22})^2 + 4J_{12}^2}\right]$.

The dominant along-fibre orientation (second output `theta`) is in $[-\pi/2, \pi/2]$ radians from horizontal.

Usage as a post-weight

```
pST.sigmaGrad = 1.5; pST.sigmaInt = 5; pST.normalize = true;
[C, theta] = structureTensorEnhance(I, pST);

R_rod    = rodGranulometryEnhance(I) .* C;      % suppress
    puncta
R_puncta = logEnhance(I)             .* (1-C);  % suppress rods
```

Parameters

sigmaGrad
Gaussian pre-smoothing before gradient (px). Default `1.5`.
Should suppress pixel noise without blurring structure boundaries.

sigmaInt
Integration scale (px). Default `5`.
Should match roughly half the structure width.

normalize Default `true`.

4.2 `orientedGaussSmooth`

Orientation-adaptive smoothing is a single-pass, non-iterative approximation to Coherence-Enhancing Diffusion [2]. The image is pre-convolved with N Gaussian kernels elongated at discrete orientations θ_k:

$$G_k(I) = I * \mathcal{N}\left(0,\, R_{\theta_k}\, \mathrm{diag}(\sigma_{\parallel}^2, \sigma_{\perp}^2)\, R_{\theta_k}^{\top}\right), \tag{4.3}$$

and the outputs are blended per-pixel using cosine-squared weights scaled by the coherence index:

$$w_k = C\cos^2(\theta - \theta_k) + \frac{1-C}{N}, \qquad \hat{I} = \frac{\sum_k w_k G_k(I)}{\sum_k w_k}. \tag{4.4}$$

Because $\sum_k \cos^2(\theta - \theta_k) = N/2$ exactly for uniform spacing, the denominator simplifies to $1 + C(N/2 - 1)$—no per-pixel loop is required. Where $C \approx 0$ (puncta, noise, flat regions) the filter reduces to isotropic averaging with scale $\sigma_{\perp}$; where $C \approx 1$ it extends $\sigma_{\parallel}$ along the fibre with minimal blurring across it.

Parameters

```
pOGS.sigmaAlong  = 4;   pOGS.sigmaAcross = 1.5;
pOGS.orientations = 8;
Ism = orientedGaussSmooth(I, pOGS);
R   = rodGranulometryEnhance(Ism, pRod);   % enhanced input
```

sigmaAlong
Along-fibre smoothing (px). Default `4`.

sigmaAcross
Across-fibre smoothing (px). Default `1.5`.
Keep small to preserve cross-fibre boundaries.

orientations
Discrete angles in $[0°, 180°)$. Default `8`.

sigmaGrad, sigmaInt
Internal structure tensor scales. Defaults `1.5` and `5`.

5
Demo Scripts

5.1 `demoMitoEnhance`

This script runs all six enhancers on both a synthetic test image (generated by `makeMitoTestImage`) and a real confocal image (`mitImage.mat`), producing eight figures.

Figure 5.1 Annotated synthetic test image with region labels (isolated rods, clustered rods, two network motifs).

Figure 5.2 Six-panel comparison on the synthetic image: raw, LoG, fiberEnhance, capsule single, capsule DoC, granulometry.

Figure 5.3 Same six-panel layout on the real confocal image.

Figure 4–6 Additional diagnostic comparisons (DoC vs single, granulometry vs LoG).

Figure 5.4 Structure tensor coherence: coherence map C, DoC$\times C$ (rod channel), and LoG$\times(1-C)$ (puncta channel).

Figure 5.5 Rod vs disk granulometry: line openings suppress puncta that disk openings preserve.

5.2 `demoPrefilter`

This script compares three pre-filtering strategies and their effect on the two most powerful rod detectors (Sections 3.1 and 3.3). One figure is generated per method; each is a 2×3 panel (rows: synthetic / real; columns: pre-filtered image, rod granulometry, fiberEnhance).

Figure 5.6 Baseline: no pre-filtering.

Figure 5.7 Perona-Malik anisotropic diffusion (`imdiffusefilt`, 15 iterations, threshold 0.03). Smooths within regions; stops at intensity edges; not directionally steered.

Figure 5.8 Guided filter (`imguidedfilter`, neighbourhood 7, regularisation 5×10^{-3}). Intensity-similarity gated; very fast; not directionally steered.

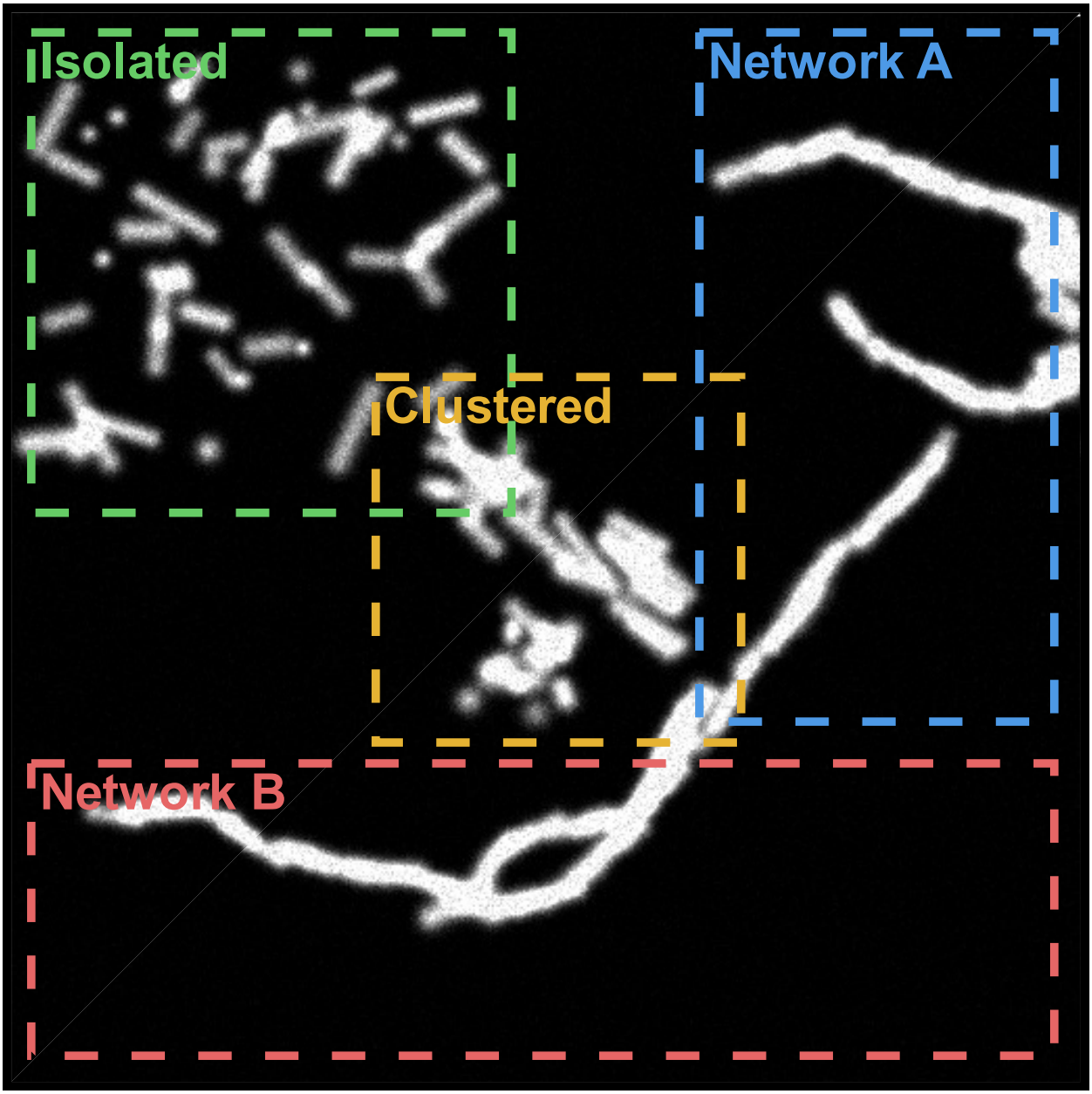

Figure 5.1: Annotated synthetic test image. Four region types are highlighted: isolated rods (green), clustered rods (amber), two network motifs (blue, red).

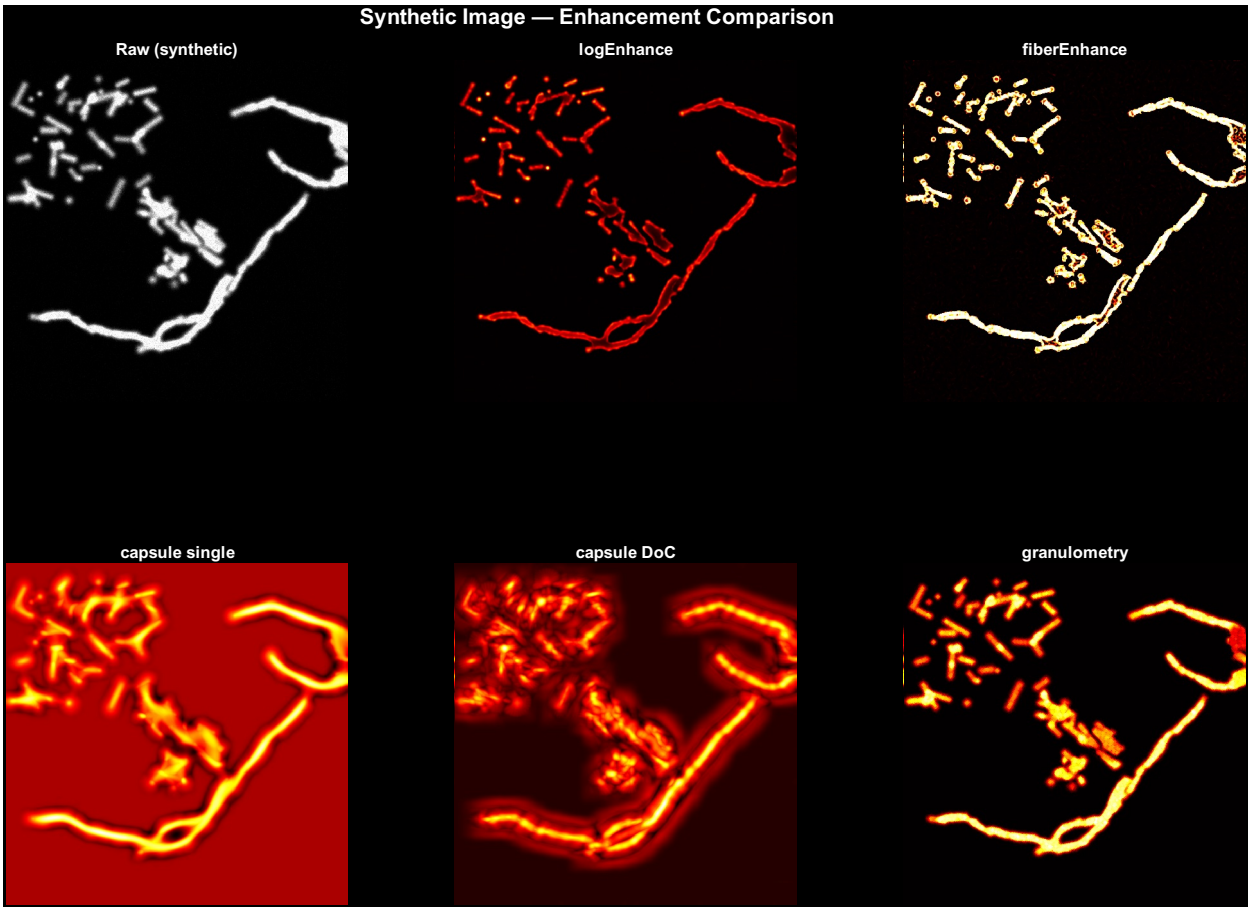

Figure 5.2: Enhancement comparison on the synthetic image. All maps are normalised to $[0, 1]$ and displayed with the *hot* colourmap.

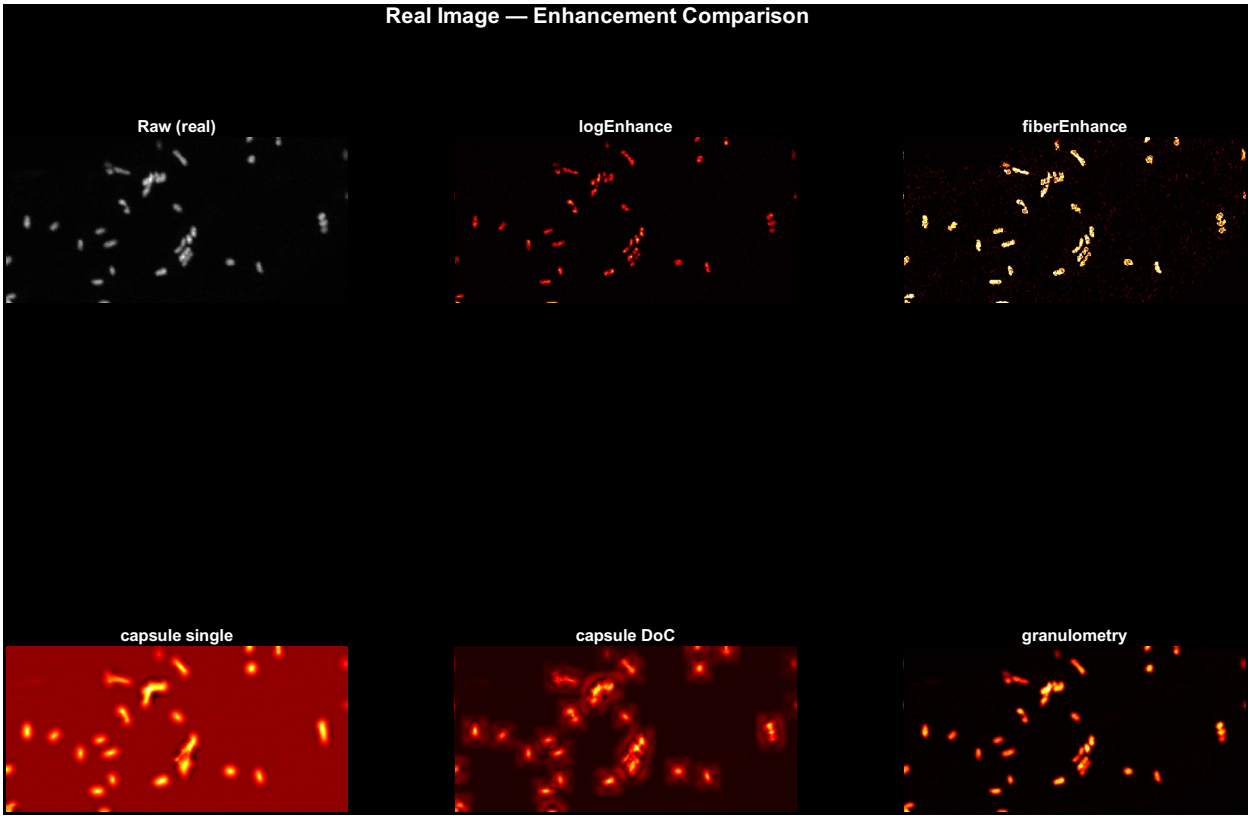

Figure 5.3: Enhancement comparison on the real confocal image.

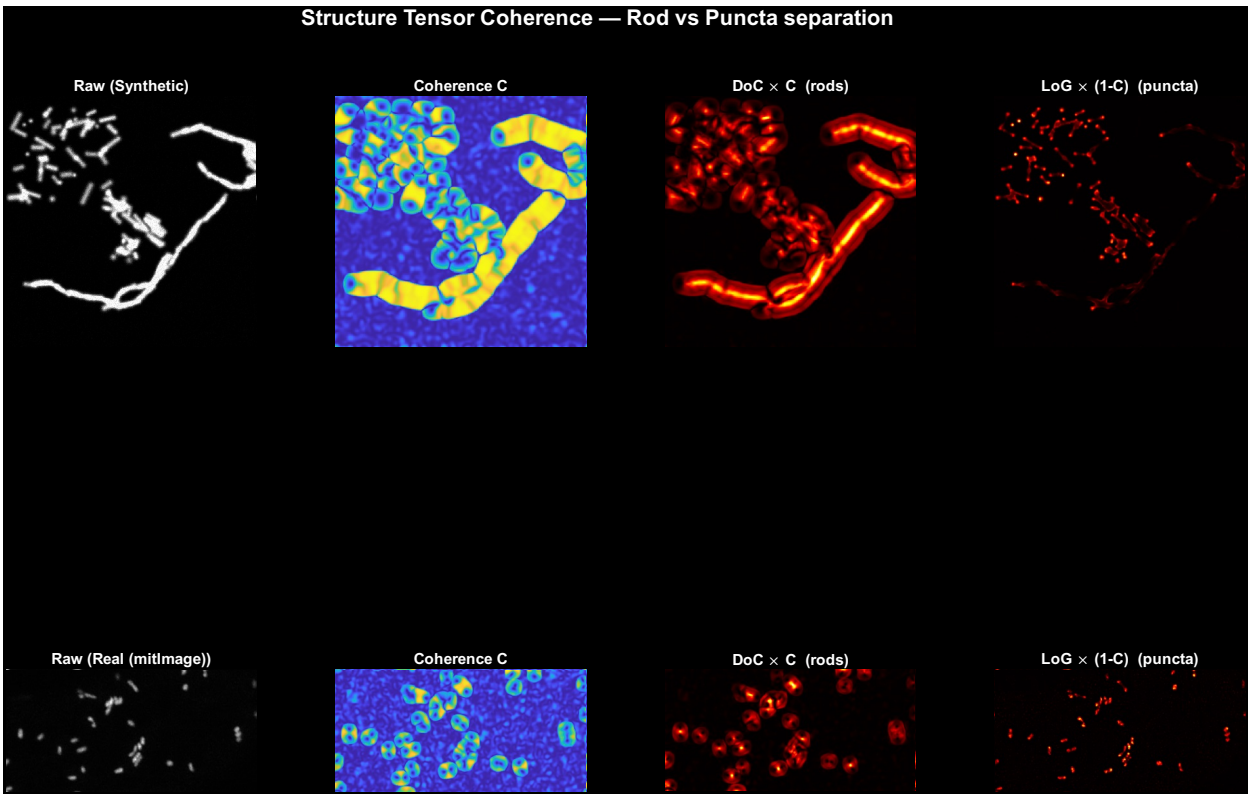

Figure 5.4: Structure tensor coherence applied as a multiplicative post-weight. Left: coherence map C (*parula* colourmap). Centre: DoC$\times C$ (rod channel). Right: LoG$\times(1-C)$ (puncta channel).

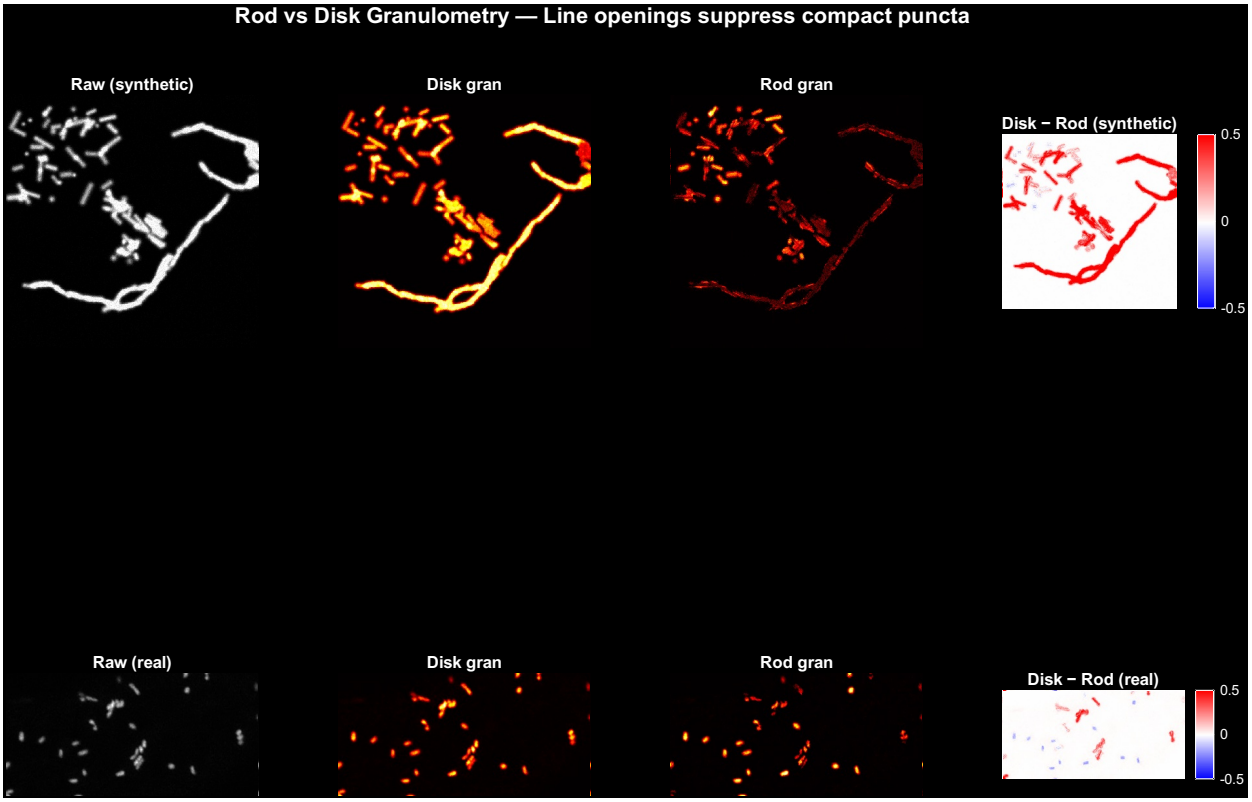

Figure 5.5: Rod vs disk granulometry. The difference map (disk−rod) reveals compact structures that disk openings preserve but line openings suppress.

Figure 5.9 Orientation-adaptive Gaussian smoothing (`orientedGaussSmooth`, $\sigma_{\parallel} = 4$, $\sigma_{\perp} = 1.5$). Directionally steered; suppresses cross-fibre noise while preserving cross-fibre boundaries.

5.3 `demoMitoEnhance` — *Cellpose comparison*

When the Cellpose add-on is installed, `demoMitoEnhance` produces a ninth figure comparing the `cyto3` segmentation with the raw image. Three columns are shown per image row:

Raw Original grayscale image (grey colourmap).

Binary mask The $\{0, 1\}$ output `R` from `cellposeEnhance` (*hot* colourmap).

Instance labels The `uint16` label image `L` rendered with `label2rgb`: each object receives a distinct hue, making touching objects visually distinguishable.

Figure not available: run `docs/exportFigures.m` with the Cellpose add-on installed to generate `fig10_cellpose.pdf`.

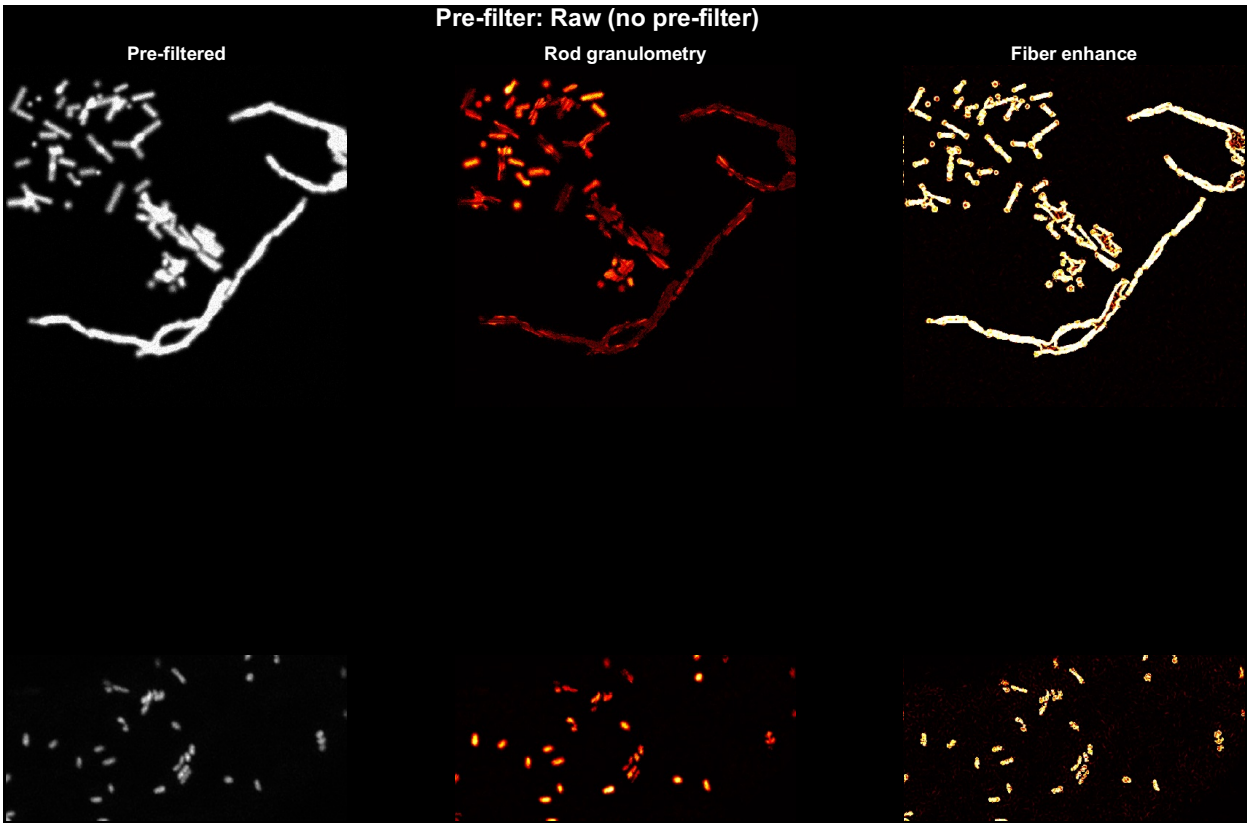

Figure 5.6: Pre-filter baseline (no pre-filtering). Top row: synthetic image. Bot-

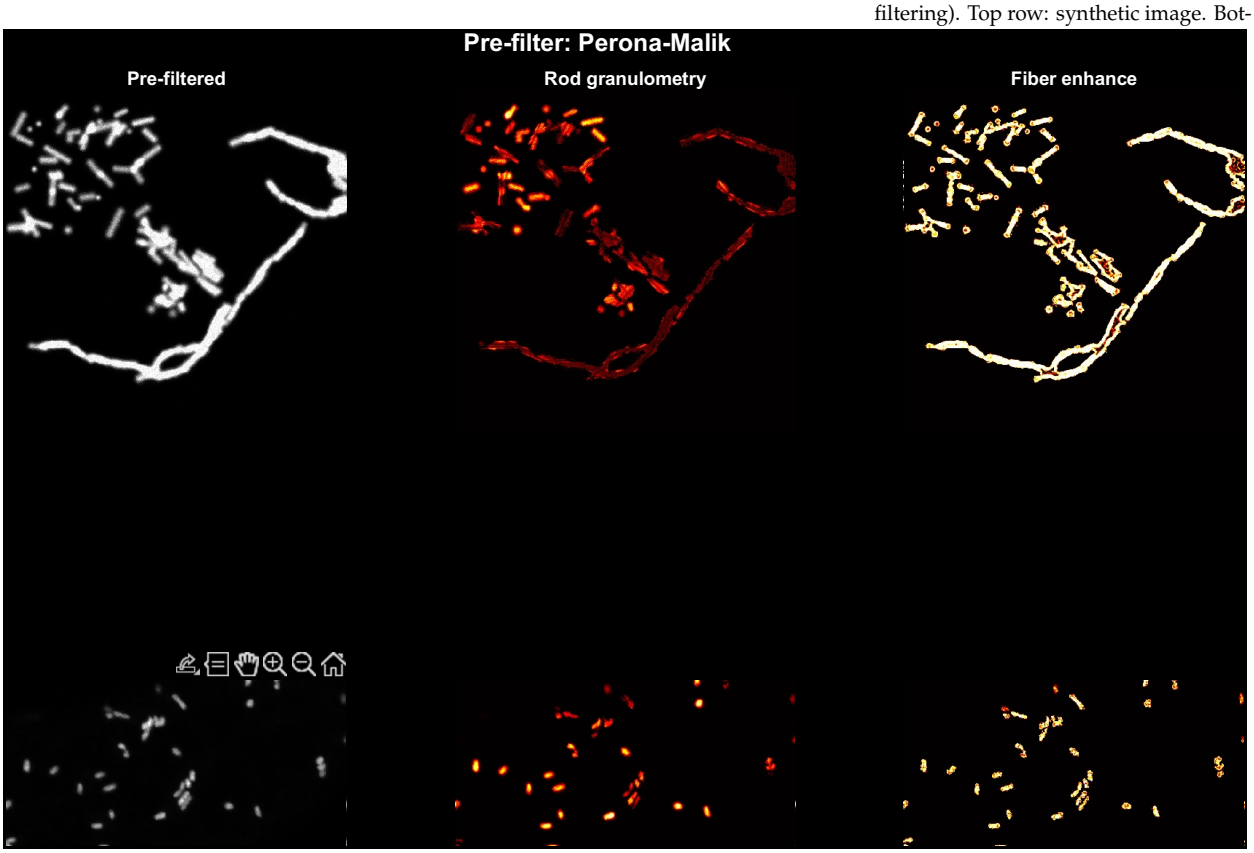

Figure 5.7: Perona-Malik anisotropic diffusion as pre-filter.

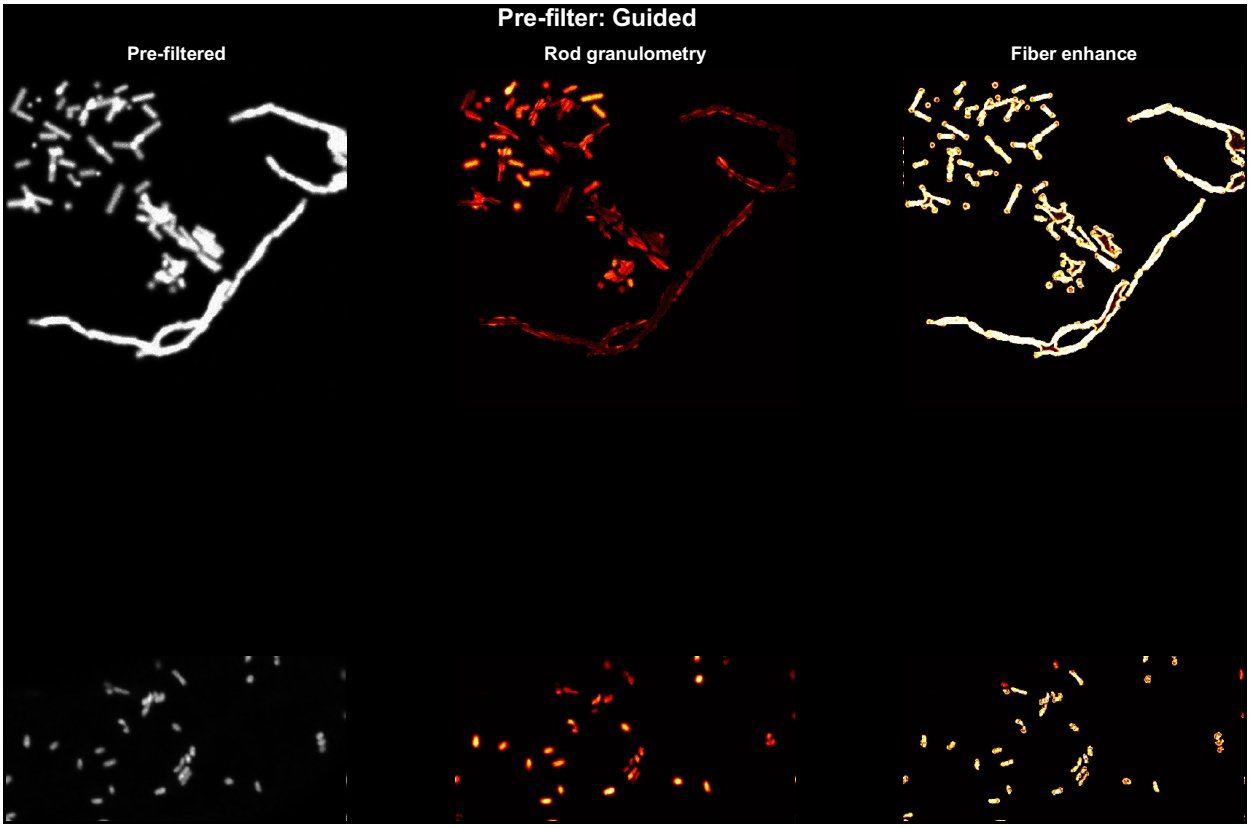

Figure 5.8: Guided filter as pre-filter.

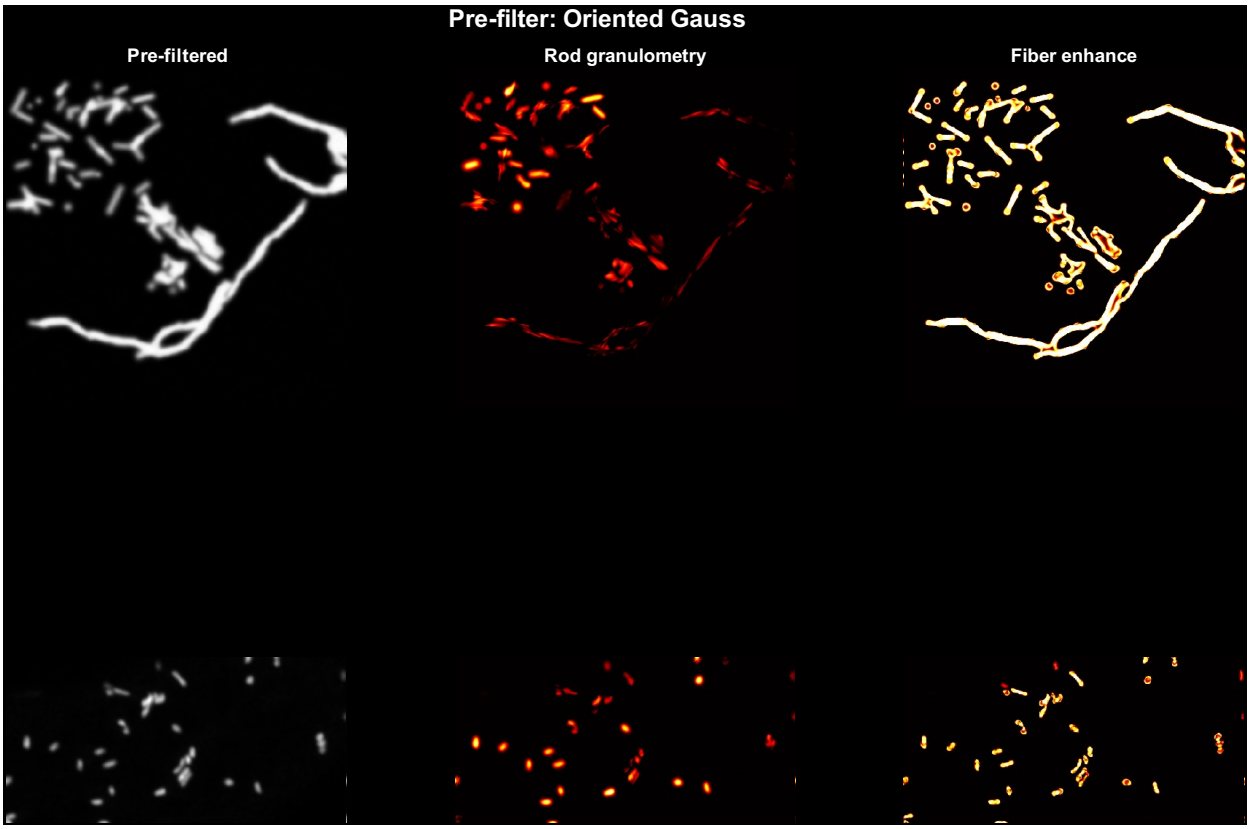

Figure 5.9: Orientation-adaptive Gaussian smoothing as pre-filter.

6
Unit Tests

A MATLAB UNIT TEST CLASS `tests/testBlobFilters.m` verifies all seven functions using the `matlab.unittest` framework introduced in R2013b. Run the full suite with:

```
results = runtests('tests/testBlobFilters');
table(results)
```

The 49 tests cover four categories for each function:

1. **Contract** – output size matches input; class is `single`; range is $[0, 1]$ when `normalize=true`.
2. **Uniform image** – constant input yields zero (or constant) output, confirming that the function does not hallucinate structure.
3. **Selectivity** – response is higher in a known synthetic structure than in a background region.
4. **Error handling** – invalid inputs and edge cases raise the expected warnings or errors.

References

7
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